

MAT1252

Mathematics for Computing

Lecture 02

Symbolic Logic

Outline

1. Today's lecture objectives
2. Propositions, connectives
3. Compound propositions example
4. Truth tables for compound propositions
5. Tautologies and contradictions
6. Logical equivalence
7. De Morgan Laws and the Laws of Propositional Logic
8. Conditional statements and contrapositives
9. Arguments

Today's Lecture Objectives

After completing this section, students should be able to:

- translate from a verbal statement into a symbolic statement, and vice versa
- construct truth tables for logical expressions involving $\wedge$, $\vee$, $\sim$ and $\rightarrow$
- use truth tables to establish logical equivalence of expressions
- use the distributive laws and de Morgan's laws
- find the negations of statements involving $\vee$ and $\wedge$
- find the negations of conditional statements
- use a truth table to establish the validity of an argument

Propositions, Connectives

A **proposition** is a statement that is either **true** or **false**.

Here are some propositions:

The file is being printed.

The system is ready.

The red light is on.

Simple representation:

p: The file is being printed.

q: The system is ready.

r: The red light is on.

Propositions, Connectives

A **proposition** is a statement that is either **true** or **false**.

Here are some propositions:

The file is being printed.

The system is ready.

The red light is on.

Simple representation:

p: The file is being printed.

q: The system is ready.

r: The red light is on.

A **compound proposition** is built from simple propositions using **connectives** (sometimes also called "operators") including:

- Not $\sim$
- And $\wedge$
- Or $\vee$
- If then $\rightarrow$

Hence, symbolic logic translates verbal sentences into symbolic expressions to make relationships easier to understand.

Propositions, Connectives

Examples of compound propositions:

- p: The file is being printed.
- q: The system is ready.
- r: The red light is on.

If the system is ready and the red light is on then the file is printed.

If the file is not printed then either the red light is not on or the system is not ready.

Either the red light is on and the file is printed or the system is not ready.

Propositions, Connectives

Examples of compound propositions:

p: The file is being printed.

q: The system is ready.

r: The red light is on.

If the system is ready and the red light is on **then** the file is printed.

If the file is not printed then either the red light is not on or the system is not ready.

Either the red light is on and the file is printed or the system is not ready.

Propositions, Connectives

Examples of compound propositions:

p: The file is being printed.

q: The system is ready.

r: The red light is on.

If the system is ready and the red light is on then the file is printed.

$$(q \wedge r) \rightarrow p$$

If the file is not printed then either the red light is not on or the system is not ready.

Either the red light is on and the file is printed or the system is not ready.

Propositions, Connectives

Examples of compound propositions:

p: The file is being printed.

q: The system is ready.

r: The red light is on.

If the system is ready and the red light is on then the file is printed.

$$(q \wedge r) \rightarrow p$$

If the file is not printed then either the red light is not on or the system is not ready.

$$\sim p \rightarrow (\sim r \vee \sim q)$$

Either the red light is on and the file is printed or the system is not ready.

Propositions, Connectives

Examples of compound propositions:

p: The file is being printed.
 q: The system is ready.
 r: The red light is on.

If the system is ready and the red light is on then the file is printed.

$$(q \wedge r) \rightarrow p$$

If the file is not printed then either the red light is not on or the system is not ready.

$$\sim p \rightarrow (\sim r \vee \sim q)$$

Either the red light is on and the file is printed or the system is not ready.

$$(r \wedge p) \vee \sim q$$

Compound propositions example

Define the following propositions:

p: Peter is driving his own car.

a: Andrew is late.

m: Max has caught the bus.

Translate a compound proposition below into symbolic notation:

Either Peter is driving his own car and Andrew is late or Max has not caught the bus.

Compound propositions example

Define the following propositions:

p : Peter is driving his own car.

a : Andrew is late.

m : Max has caught the bus.

Translate a compound proposition below into a symbolic notation:

Either (Peter is driving his own car and Andrew is late) or Max has not caught the bus.

$$(p \wedge a) \vee \sim m$$

Compound propositions example

Define the following propositions:

p: Peter is driving his own car.

a: Andrew is late.

m: Max has caught the bus.

Translate into simple English:

$$m \wedge (\sim p \vee \sim a)$$

Compound propositions example

Define the following propositions:

p : Peter is driving his own car.

a : Andrew is late.

m : Max has caught the bus.

Translate into simple English:

$$m \wedge (\sim p \vee \sim a)$$

Max has caught the bus and either Peter is **not** driving his own car or Andrew is **not** late.

Compound propositions example

Define the following propositions:

s: Sue is a geologist.

a: Anne sells insurance.

m: Mary teaches music.

Write each of the following in symbols:

If Mary teaches music then Sue is a geologist and also Anne does not sell insurance.

Compound propositions example

Define the following propositions:

s: Sue is a geologist.

a: Anne sells insurance.

m: Mary teaches music.

Write each of the following in symbols:

If Mary teaches music then Sue is a geologist and also Anne does not sell insurance.

$$m \rightarrow (s \wedge \sim a)$$

Compound propositions example

Translate into simple English:

$$\sim s \rightarrow (\sim m \wedge \sim a)$$

s: Sue is a geologist.

a: Anne sells insurance.

m: Mary teaches music.

Compound propositions example

Translate into simple English:

$$\sim s \rightarrow (\sim m \wedge \sim a)$$

s: Sue is a geologist.

a: Anne sells insurance.

m: Mary teaches music.

If Sue is not a geologist then Mary does not teach music and Anne does not sell insurance.

Truth Table

A **truth table** shows all the possible combinations of the simple propositions and the corresponding truth values of the compound proposition.

A **truth value** of **1** indicates that a statement is **true (T)**.

A **truth value** of **0** indicates that a statement is **false (F)**.

To be able to use **truth tables** as an important tool, we first have to define **connectives**.

The NOT connective

NOT - the simplest operation denoted by the symbol " $\sim$ ".

$\sim p$ is read as "not p" or "It is not true that p".

For example if p means "Today is Saturday", then

$\sim p$ means "It is not true that today is Saturday",

or in simpler English, "Today is not Saturday."

The NOT connective

NOT - the simplest operation denoted by the symbol " $\sim$ ".

$\sim p$ is read as "not p" or "It is not true that p".

For example if p means "Today is Saturday", then

$\sim p$ means "It is not true that today is Saturday",

or in simpler English, "Today is not Saturday."

NOT Truth Table

p	$\sim p$
0	1
1	0

Note: Exactly ONE of p and $\sim p$ is true.

The NOT connective

NOT - the simplest operation denoted by the symbol " $\sim$ ".

$\sim p$ is read as "not p" or "It is not true that p".

For example if p means "Today is Saturday", then

$\sim p$ means "It is not true that today is Saturday",

or in simpler English, "Today is not Saturday."

NOT Truth Table

p	$\sim p$
0	1
1	0

For example:

p: $3 + 4 = 7$

q: It is raining

r: $8 > 15$

$\sim p$: $3 + 4 \neq 7$

$\sim q$: It is not raining.

$\sim r$: $8 \leq 15$

Note: Exactly ONE of p and $\sim p$ is true.

The AND connective

AND $p \wedge q$ is read as "**p and q**"
 "AND" carries with it the expectation of both.

AND Truth Table

p	q	$p \wedge q$
0	0	0
0	1	0
1	0	0
1	1	1

An **AND** statement is true if **both parts are true**; otherwise it is false.

The OR connective

OR $p \vee q$ is read as "p or q"

An OR statement is true if **at least one part is true**, and it is also true if both parts are true.

OR Truth Table

p	q	$p \vee q$
0	0	0
0	1	1
1	0	1
1	1	1

An OR statement is false if both parts are false; otherwise it is true.

The OR connective

OR $p \vee q$ is read as "p or q"

An OR statement is true if **at least one part is true**, and it is also true if both parts are true.

OR Truth Table

p	q	$p \vee q$
0	0	0
0	1	1
1	0	1
1	1	1

Note:

OR is different to XOR ("exclusive or"). An XOR statement is true if **exactly one part is true**.

An OR statement is false if both parts are false; otherwise it is true.

Truth tables for compound propositions

Let us construct the truth table for $p \wedge (q \vee \sim r)$

Step 1 Build the table and list all the possible combinations for the number of existing propositions

p	q	r	$\sim r$	$q \vee \sim r$	$p \wedge (q \vee \sim r)$
0	0	0			
0	0	1			
0	1	0			
0	1	1			
1	0	0			
1	0	1			
1	1	0			
1	1	1			

Truth tables for compound propositions

Let us construct the truth table for $p \wedge (q \vee \sim r)$

Step 1 Build the table and list all the possible combinations for the number of existing propositions

p	q	r	$\sim r$	$q \vee \sim r$	$p \wedge (q \vee \sim r)$
0	0	0			
0	0	1			
0	1	0			
0	1	1			
1	0	0			
1	0	1			
1	1	0			
1	1	1			

Step 2 Fill column by column according to relation defined for the particular column

p	q	r	$\sim r$	$q \vee \sim r$	$p \wedge (q \vee \sim r)$
0	0	0	1		
0	0	1	0		
0	1	0	1		
0	1	1	0		
1	0	0	1		
1	0	1	0		
1	1	0	1		
1	1	1	0		

p	$\sim p$
0	1
1	0

Truth tables for compound propositions

Let us construct the truth table for $p \wedge (q \vee \sim r)$

Step 3 Continue filling columns

p	q	r	$\sim r$	$q \vee \sim r$	$p \wedge (q \vee \sim r)$
0	0	0	1	1	
0	0	1	0	0	
0	1	0	1	1	
0	1	1	0	1	
1	0	0	1	1	
1	0	1	0	0	
1	1	0	1	1	
1	1	1	0	1	

p	q	$p \vee q$
0	0	0
0	1	1
1	0	1
1	1	1

Truth tables for compound propositions

Let us construct the truth table for $p \wedge (q \vee \sim r)$

Step 3 Continue filling columns

p	q	r	$\sim r$	$q \vee \sim r$	$p \wedge (q \vee \sim r)$
0	0	0	1	1	
0	0	1	0	0	
0	1	0	1	1	
0	1	1	0	1	
1	0	0	1	1	
1	0	1	0	0	
1	1	0	1	1	
1	1	1	0	1	

p	q	$p \vee q$
0	0	0
0	1	1
1	0	1
1	1	1

Step 4 Fill final column

p	q	r	$\sim r$	$q \vee \sim r$	$p \wedge (q \vee \sim r)$
0	0	0	1	1	0
0	0	1	0	0	0
0	1	0	1	1	0
0	1	1	0	1	0
1	0	0	1	1	1
1	0	1	0	0	0
1	1	0	1	1	1
1	1	1	0	1	1

p	q	$p \wedge q$
0	0	0
0	1	0
1	0	0
1	1	1

Truth tables example

Construct the truth table for the following proposition:

$$((p \wedge \sim q) \rightarrow r) \wedge (\sim p \rightarrow \sim r)$$

Truth tables example

Construct the truth table for the following proposition:

$$((p \wedge \sim q) \rightarrow r) \wedge (\sim p \rightarrow \sim r)$$

p	q	r	$\sim q$	$p \wedge \sim q$	$(p \wedge \sim q) \rightarrow r$	$\sim p$	$\sim r$	$\sim p \rightarrow \sim r$	$((p \wedge \sim q) \rightarrow r) \wedge (\sim p \rightarrow \sim r)$		
0	0	0	1	0	1	1	1	1		1	
0	0	1	1	0	1	1	0	0		0	
0	1	0	0	0	1	1	1	1		1	
0	1	1	0	0	1	1	0	0		0	
1	0	0	1	1	0	0	1	1		0	
1	0	1	1	1	1	0	0	1		1	
1	1	0	0	0	1	0	1	1		1	
1	1	1	0	0	1	0	0	1		1	

Tautologies and contradictions

A proposition that is **true for every combination of truth values** is called a **tautology**.

Tautologies and contradictions

A proposition that is **true for every combination of truth values** is called a **tautology**.

“Julia is here, or Julia is not here.”

Tautologies and contradictions

A proposition that is **true for every combination of truth values** is called a **tautology**.

Tautologies and contradictions

A proposition that is **true for every combination of truth values** is called a **tautology**.

Claim: The proposition $(p \vee q) \vee (\sim p \wedge \sim q)$ is a tautology

Step 1

p	q	$p \vee q$	$\sim p \wedge \sim q$	$(p \vee q) \vee (\sim p \wedge \sim q)$
0	0			
0	1			
1	0			
1	1			

Tautologies and contradictions

A proposition that is **true for every combination of truth values** is called a **tautology**.

Claim: The proposition $(p \vee q) \vee (\sim p \wedge \sim q)$ is a tautology

Step 1

p	q	$p \vee q$	$\sim p \wedge \sim q$	$(p \vee q) \vee (\sim p \wedge \sim q)$
0	0			
0	1			
1	0			
1	1			

Final step

p	q	$p \vee q$	$\sim p \wedge \sim q$	$(p \vee q) \vee (\sim p \wedge \sim q)$
0	0	0	1	1
0	1	1	0	1
1	0	1	0	1
1	1	1	0	1

Tautologies and contradictions

A proposition that is **false for every possible combination of truth values** is called a **contradiction**.

Tautologies and contradictions

A proposition that is **false for every possible combination of truth values** is called a **contradiction**.

“Julia is here and Julia is not here.”

Tautologies and contradictions

A proposition that is **false for every possible combination of truth values** is called a **contradiction**.

Tautologies and contradictions

A proposition that is **false for every possible combination of truth values** is called a **contradiction**.

Claim: The proposition $(p \wedge q) \wedge (\sim p \vee \sim q)$ is a contradiction.

Step 1

p	q	$p \wedge q$	$\sim p \vee \sim q$	$(p \wedge q) \wedge (\sim p \vee \sim q)$
0	0			
0	1			
1	0			
1	1			

Tautologies and contradictions

A proposition that is **false for every possible combination of truth values** is called a **contradiction**.

Claim: The proposition $(p \wedge q) \wedge (\sim p \vee \sim q)$ is a contradiction.

Step 1

p	q	$p \wedge q$	$\sim p \vee \sim q$	$(p \wedge q) \wedge (\sim p \vee \sim q)$
0	0			
0	1			
1	0			
1	1			

Final step

p	q	$p \wedge q$	$\sim p \vee \sim q$	$(p \wedge q) \wedge (\sim p \vee \sim q)$
0	0	0	1	0
0	1	0	1	0
1	0	0	1	0
1	1	1	0	0

Logical equivalence

We often have different, but **equivalent**, logical expressions.

To determine whether two given propositions have exactly the same logical meaning we use a truth table to check it.

Equivalent statements have the same truth table.

Logical equivalence is denoted by the symbol $\equiv$.

Show that $\sim p \wedge (p \vee \sim q) \equiv \sim (p \vee q)$

LHS					RHS		
p	q	$\sim p$	$\sim q$	$p \vee \sim q$	$\sim p \wedge (p \vee \sim q)$	$p \vee q$	$\sim (p \vee q)$
0	0						
0	1						
1	0						
1	1						

Logical equivalence

Show that $\sim p \wedge (p \vee \sim q) \equiv \sim(p \vee q)$

p	q	$\sim p$	$\sim q$	$p \vee \sim q$	$\sim p \wedge (p \vee \sim q)$	$p \vee q$	$\sim(p \vee q)$
0	0	1	1	1	1		
0	1	1	0	0	0		
1	0	0	1	1	0		
1	1	0	0	1	0		

p	q	$\sim p$	$\sim q$	$p \vee \sim q$	$\sim p \wedge (p \vee \sim q)$	$p \vee q$	$\sim(p \vee q)$
0	0	1	1	1	1	0	1
0	1	1	0	0	0	1	0
1	0	0	1	1	0	1	0
1	1	0	0	1	0	1	0

Two given propositions $\sim p \wedge (p \vee \sim q) \equiv \sim(p \vee q)$ are logically equivalent.

Truth tables example

Use a truth table to show that :

$$p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$$

Truth tables example

Use a truth table to show that :

$$p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$$

p	q	r	$q \vee r$	$p \wedge (q \vee r)$		$p \wedge q$	$p \wedge r$	$(p \wedge q) \vee (p \wedge r)$	
0	0	0	0		0	0	0		0
0	0	1	1		0	0	0		0
0	1	0	1		0	0	0		0
0	1	1	1		0	0	0		0
1	0	0	0		0	0	0		0
1	0	1	1		1	0	1		1
1	1	0	1		1	1	0		1
1	1	1	1		1	1	1		1

De Morgan's laws

De Morgan's laws show how to negate statements that contain an AND or an OR.

Short story:

Negation of AND $\longrightarrow$ OR

Negation of OR $\longrightarrow$ AND

The negation of an AND statement

One of de Morgan's laws defines the negation of an AND statement:

$$\text{NOT } (p \text{ AND } q) \equiv \text{NOT } p \text{ OR NOT } q$$

$$\sim(p \wedge q) \equiv \sim p \vee \sim q$$

p	q	$p \wedge q$	$\sim(p \wedge q)$	$\sim p$	$\sim q$	$\sim p \vee \sim q$
0	0	0		1	1	
0	1	0		1	0	
1	0	0		0	1	
1	1	1		0	0	

p	q	$p \wedge q$	$\sim(p \wedge q)$	$\sim p$	$\sim q$	$\sim p \vee \sim q$
0	0	0	1	1	1	1
0	1	0	1	1	0	1
1	0	0	1	0	1	1
1	1	1	0	0	0	0

The negation of an OR statement

The second of de Morgan's laws:

$$\text{NOT}(p \text{ OR } q) \equiv \text{NOT } p \text{ AND NOT } q$$

$$\sim(p \vee q) \equiv \sim p \wedge \sim q$$

p	q	$p \vee q$	$\sim(p \vee q)$	$\sim p$	$\sim q$	$\sim p \wedge \sim q$
0	0	0		1	1	
0	1	1		1	0	
1	0	1		0	1	
1	1	1		0	0	

p	q	$p \vee q$	$\sim(p \vee q)$	$\sim p$	$\sim q$	$\sim p \wedge \sim q$
0	0	0	1	1	1	1
0	1	1	0	1	0	0
1	0	1	0	0	1	0
1	1	1	0	0	0	0

Compound propositions example

Use de Morgan's laws to find the negation of each of the following statements. Give your answers in simple English.

Peter has long hair and he is **not** wearing a coat.

David plays guitar and likes tennis.

Compound propositions example

Use de Morgan's laws to find the negation of each of the following statements. Give your answers in simple English.

Peter has long hair and he is **not** wearing a coat.

Peter does **not** have long hair or he is wearing a coat.

David plays guitar and likes tennis.

Compound propositions example

Use de Morgan's laws to find the negation of each of the following statements. Give your answers in simple English.

Peter has long hair and he is **not** wearing a coat.

Peter does **not** have long hair or he is wearing a coat.

David plays guitar and likes tennis.

David does **not** play guitar or he does **not** like tennis.

The Laws of Propositional Logic

We frequently need to simplify logical expressions or to check whether given logical expressions are logically equivalent.

One way to do these things is to use the laws of logic.

For our purposes the most important laws are the **distributive laws** and **de Morgan's laws**.

Distributive Laws

$$p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$$

$$p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$$

De Morgan's Laws

$$\sim(p \vee q) \equiv \sim p \wedge \sim q$$

$$\sim(p \wedge q) \equiv \sim p \vee \sim q$$

The Laws of Propositional Logic

We frequently need to simplify logical expressions or to check whether given logical expressions are logically equivalent.

One way to do these things is to use the laws of logic.

For our purposes the most important laws are the **distributive laws** and **de Morgan's laws**.

Distributive Laws

$$p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$$

$$p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$$

De Morgan's Laws

$$\sim(p \vee q) \equiv \sim p \wedge \sim q$$

$$\sim(p \wedge q) \equiv \sim p \vee \sim q$$

A bit like when we do

$$2 \times (4 + 5) = (2 \times 4) + (2 \times 5)$$

Conditional statements

In computing we often use statements of the form "if ... then ..." to mean that if particular conditions are satisfied then certain consequences should follow.

A proposition of the form "if p then q" is called a **conditional statement**, and is represented by $p \rightarrow q$.

The symbolic statement is usually read as "if p then q" or as "p implies q".

Truth table:

p	q	$p \rightarrow q$
0	0	1
0	1	1
1	0	0
1	1	1

Conditional statements

In computing we often use statements of the form "If ... then ..." to mean that if particular conditions are satisfied then certain consequences should follow.

A proposition of the form "If p then q " is called a **conditional statement**, and is represented by $p \rightarrow q$.

The symbolic statement is usually read as "if p then q " or as " p implies q ".

Truth table:

p	q	$p \rightarrow q$	$\sim p$	$\sim p \vee q$
0	0	1	1	1
0	1	1	1	1
1	0	0	0	0
1	1	1	0	1

Notice that $p \rightarrow q$ is logically equivalent to $\sim p \vee q$.

Conditional statements

Example

p: “I live in Perth”

q: “I live in Western Australia”

p	q	$p \rightarrow q$	Interpretation
0	0	1	“It is possible that I don’t live in Perth and I don’t live in Western Australia”
0	1	1	“It is possible that I don’t live in Perth but I do live in Western Australia.”
1	0	0	“It is impossible that I live in Perth but I do not live in Western Australia.”
1	1	1	“It is possible that I live in Perth and I live in Western Australia.”

The negation of a conditional statement

The negation of the conditional $p \rightarrow q$ is:

$$\sim(p \rightarrow q) \equiv p \wedge \sim q$$

p	q	$p \rightarrow q$	$\sim(p \rightarrow q)$	p	$\sim q$	$p \wedge \sim q$
0	0	1	0	0	1	0
0	1	1	0	0	0	0
1	0	0	1	1	1	1
1	1	1	0	1	0	0

Note that we can derive the result from De Morgan's laws, since

$$\sim(p \rightarrow q) \equiv \sim(\underbrace{\sim p \vee q}_{\text{from last slide}}) \equiv (\underbrace{\sim \sim p}_{\text{from de Morgan}}) \wedge (\sim q) \equiv p \wedge \sim q$$

The contrapositive

The **contrapositive** of a conditional statement $p \rightarrow q$ is the statement $\sim q \rightarrow \sim p$.

The conditional $p \rightarrow q$ and its contra-positive $\sim q \rightarrow \sim p$ are logically equivalent.

$$p \rightarrow q \equiv \sim q \rightarrow \sim p$$

p	q	$p \rightarrow q$	$\sim q$	$\sim p$	$\sim q \rightarrow \sim p$
0	0	1	1	1	1
0	1	1	0	1	1
1	0	0	1	0	0
1	1	1	0	0	1

Contrapositive example

Consider the conditional statement

“If I am in Perth then I am in Western Australia.”

Contrapositive example

Consider the conditional statement

“If I am in Perth then I am in Western Australia.”

The contrapositive statement is

“If I am not in Western Australia, then I am not in Perth.”

The truth values of both statements are the same; it is impossible for one to be true and the other false.

The converse of a conditional statement

$p \rightarrow q$ is a conditional statement, its **converse** is $q \rightarrow p$.

These two statements may look similar, but there is no logical connection!

The two statements are completely independent of each other.

One of the most common errors in logic is to confuse a conditional statement and its converse.

CHALLENGE

Find examples of statements p and q such that

- $p \rightarrow q$ is true and $q \rightarrow p$ is true
- $p \rightarrow q$ is true and $q \rightarrow p$ is false

Arguments

An argument consists of a series of statements (called premises) and a conclusion such as shown below.

Notice that the premises are listed one after the other and separated from the conclusion by a horizontal or vertical line, as appropriate.

If the printer is on line and the file is
 loaded then the file will be printed.

(premise)

The file is loaded.

(premise)

The file is not being printed.

(premise)

Therefore the printer is not on line.

(conclusion)

Arguments

We write an argument symbolically as $P_1, P_2, \dots, P_n \mid C$, where $P_1, P_2, \dots, P_n$ are the premises and C is the conclusion.

To show that the argument $P_1, P_2, \dots, P_n \mid C$ is valid, we must check that whenever the premises $P_1, P_2, \dots, P_n$ are true that the conclusion C is also true.

Arguments

IF

THEN

Premises are all true $\rightarrow$ Conclusion is true

To show that an argument is **valid**, you must show that for EVERY row in the truth table where the premises are all true, that the conclusion is true.

To show that an argument is **not valid**, you must find AT LEAST ONE row in the truth table where:

- The premises are all true
- The conclusion is false.

Arguments

IF

THEN

Premises are all true $\rightarrow$ Conclusion is true

Note that:

- If the truth table has no rows where the premises are all true, then the argument is **valid**.
- In the rows of the truth table where the premises are all true, if you find that in some of them the conclusion is true and in others the conclusion is false, the argument is **not valid**.

Arguments

Example 1

Test the validity of the argument: $p \rightarrow (q \wedge \sim r), q, r \mid \sim p$

	prem		prem		prem	conc
p	q	r	$\sim r$	$q \wedge \sim r$	$p \rightarrow (q \wedge \sim r)$	$\sim p$
0	0	0	1	0	1	1
0	0	1	0	0	1	1
0	1	0	1	1	1	1
0	1	1	0	0	1	1
1	0	0	1	0	0	0
1	0	1	0	0	0	0
1	1	0	1	1	1	0
1	1	1	0	0	0	0

There is only one row in which the premises are all true, and in this row the conclusion is also true. **Therefore the argument is valid.**

Arguments

Example 2

Test the validity of the argument: $p \vee q, p \rightarrow \sim r, \sim r \mid p$

conc		prem		prem	prem	
p	q	r	$\sim r$	$p \vee q$	$p \rightarrow \sim r$	check
0	0	0	1	0	1	
0	0	1	0	0	1	
0	1	0	1	1	1	no
0	1	1	0	1	1	
1	0	0	1	1	1	yes
1	0	1	0	1	0	
1	1	0	1	1	1	yes
1	1	1	0	1	0	

There are three rows in which all the premises are true, but in one of these the conclusion is false. **Therefore the argument is not valid.**

Today's Lecture Objectives

After completing this section, students should be able to:

- translate from a verbal statement into a symbolic statement, and vice versa
- construct truth tables for logical expressions involving $\wedge$, $\vee$, $\sim$ and $\rightarrow$
- use truth tables to establish logical equivalence of expressions
- use the distributive laws and de Morgan's laws
- find the negations of statements involving $\vee$ and $\wedge$
- find the negations of conditional statements
- use a truth table to establish the validity of an argument

The End